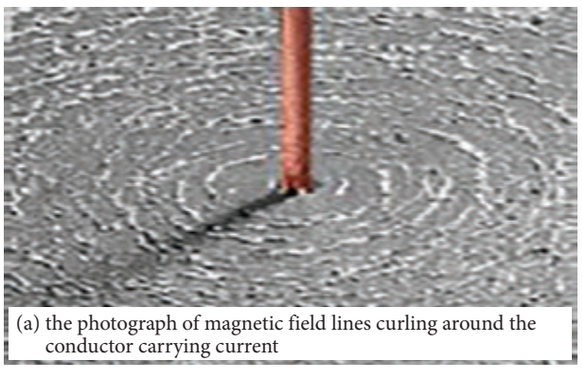
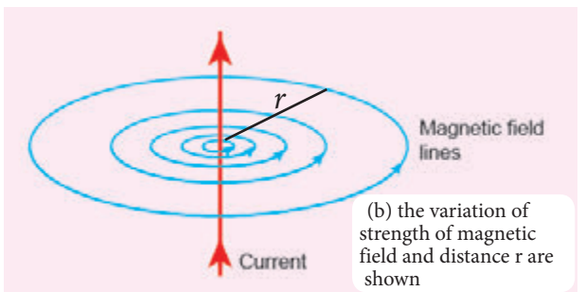
## 3.7.2 Magnetic field around a straight current carrying conductor and circular loop

### (a) Current carrying straight conductor:



(a) the photograph of magnetic field lines curling around the conductor carrying current



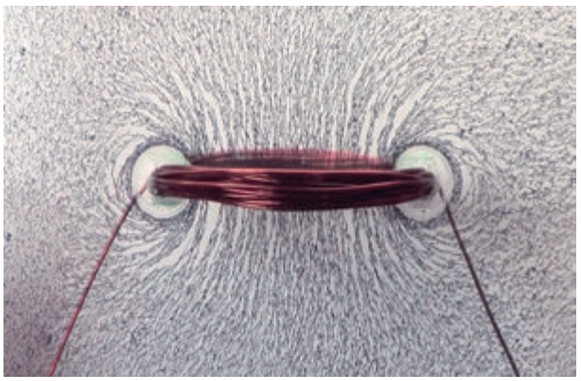

(b) the variation of strength of magnetic field and distance r are shown

**Figure 3.33** Magnetic field lines around straight, long wire carrying current

Suppose we keep a magnetic compass near a current carrying straight conductor, then the needle of the magnetic compass experiences a torque and deflects to align in the direction of the magnetic field at that point. Tracing out the direction shown by magnetic compass, we can draw the magnetic field lines at a distance. For a straight current carrying conductor, the nature of magnetic field is like concentric circles having their center at the axis of the conductor as shown in Figure 3.33 (a).

The direction of circular magnetic field lines will be clockwise or anticlockwise depending on the direction of current in the conductor. If the strength (or magnitude) of the current is increased then the density of the magnetic field will also increase. The strength of the magnetic field (B) decreases as the distance (r) from the conductor increases are shown in Figure 3.33 (b).

### (b) Circular coil carrying current



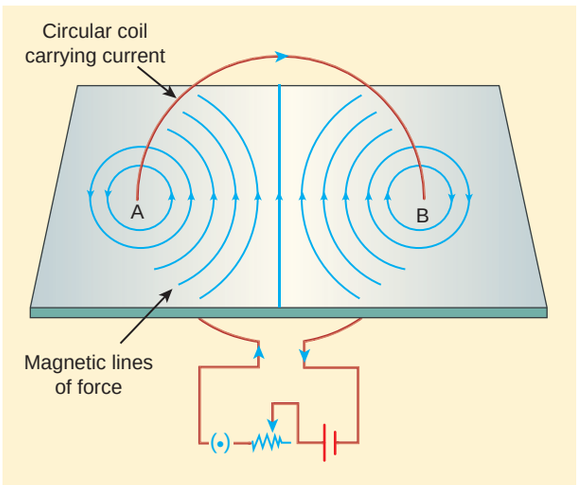



**Figure 3.34** The magnetic field lines curling around the circular coil carrying current.

Suppose we keep a magnetic compass near a current carrying circular conductor, then the needle of the magnetic compass experiences a torque and deflects to align in the direction of the magnetic field at that point. We can notice that at the points A and B in the vicinity of the coil, the magnetic field lines are circular. The magnetic field lines are nearly parallel to each other near



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